

COUCHBASE PLATFORM ARCHITECTURE

End-to-End Data Flow: Sources, Streaming, Integration & Deployment



Agenda

A single reference for how data flows from source systems into Couchbase, how the platform is structured, and how it is secured.

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Project Background

Commercial Banking Enhanced Reporting

Initiative

A Wintrust Commercial Banking – Treasury Management (TM) initiative to expand TM services to a new market segment: Large Commercial and Corporate clients. The project builds the business, technology, processing, and service capabilities needed to meet the demands of this segment.

Client Needs

- Prior-day and same-day visibility into balances and transactions
- Standard and custom reports at the BAI code level
- Cash position and cash-flow forecasting from near real-time data
- Enrichment of transaction information for context and clarity
- Transaction search across current and historical data spanning 7 years

BUSINESS HUB

Client-facing UI portal — built & operated by FIS

Gives commercial clients a single place to view balances, transactions, cash position and forecast, run standard or custom reports, and search transaction history.

DATAHUB

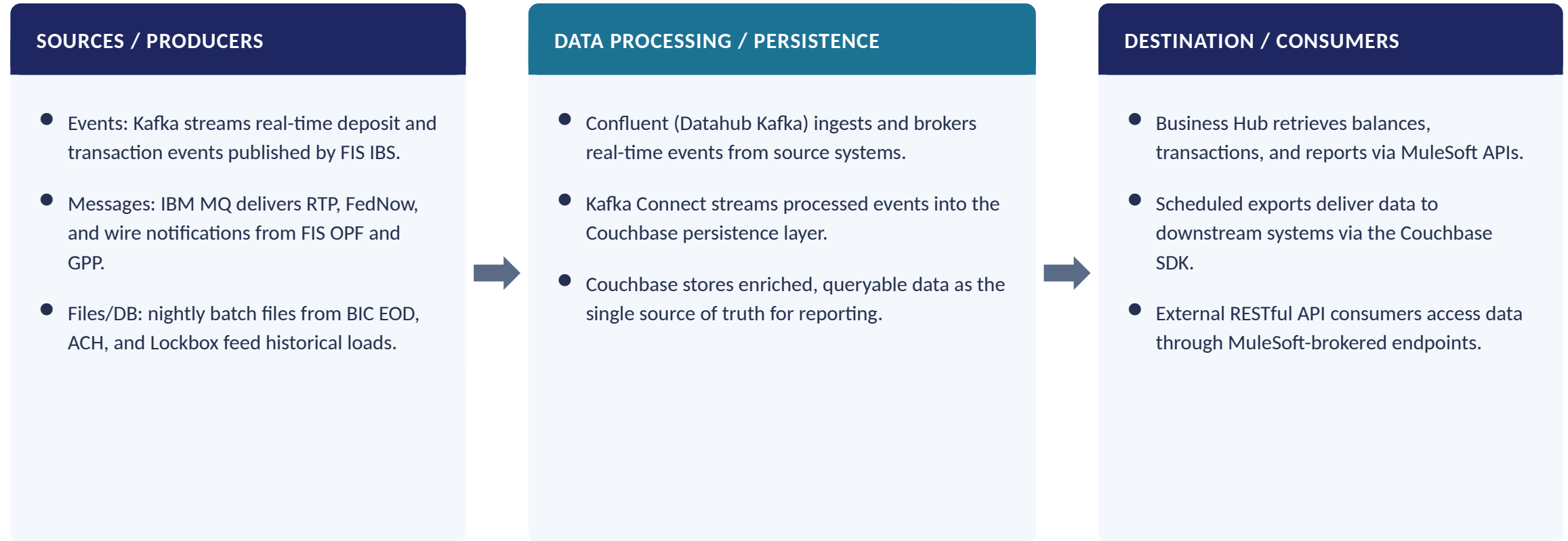
Integration & data layer — built & operated by Wintrust Enterprise DM

Acquires data from core banking and TPS systems, ingests it at near real-time and end-of-day frequencies, processes and enriches it to a standardized BAI code level, and provisions it to Business Hub.

Datahub — powered by Couchbase and the streaming/integration architecture covered in this deck — is the backbone that makes real-time, enriched, historical reporting possible.

High-Level Architecture Overview

Sources / Producers → Data Processing / Persistence → Destination / Consumers



Three ingestion patterns — real-time events, message queues, and batch files — converge in Couchbase as the central persistence layer, then fan out to consumers via MuleSoft.

Data Sources

Grouped by ingestion pattern

CORE BANKING

- IBS (FIS Core Banking): system of record for deposit accounts and core transaction processing.

REAL-TIME INTEGRATIONS

- RTP / FedNow: instant payment rails processed through FIS OPF in real time.
- Wires: inbound/outbound transfers confirmed via GPP advice messages.

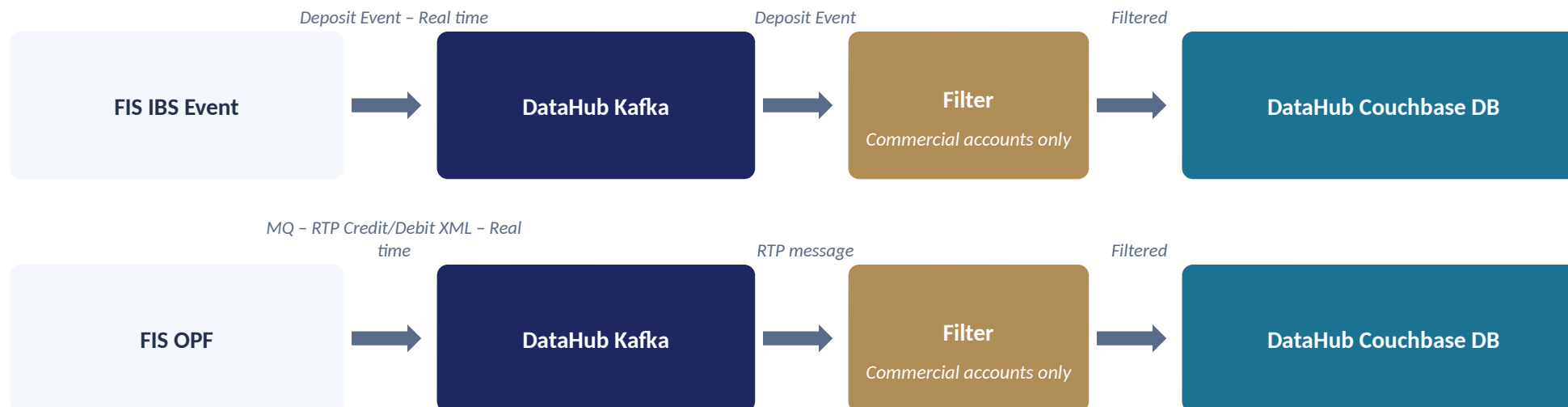
BATCH-BASED TPS

- ACH: origination, receipt, and return transactions from FIS ACH Tracker.
- Lockbox: deposit summary files produced by Deluxe, loaded nightly.
- Cards: card transaction settlement captured through core batch processing.

Sources span three velocities — **core banking (continuous)**, **real-time payment rails (instant)**, and **batch TPS (scheduled)** — all converging into a single ingestion layer before reaching Couchbase.

Streaming Layer

Real-time ingestion via Kafka — with commercial account filtering



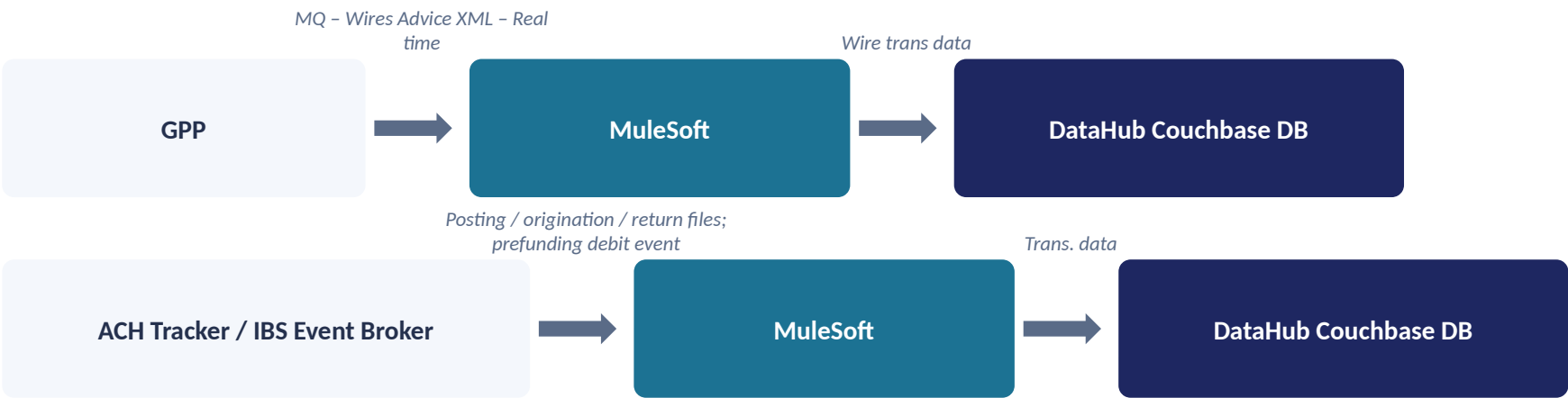
IBS Deposits (top) and RTP / FedNow (bottom) are the two sources that stream through Kafka.

Kafka isn't just a pass-through — it's the filtering checkpoint that scopes real-time events to commercial accounts before they ever reach Couchbase. Wires, ACH, and Lockbox reach Couchbase through other real-time or batch paths, covered next in Mulesoft Inbound Integration.

Mulesoft Inbound Integration

Inbound broker for Wires & ACH — universal outbound gateway to Business Hub

INBOUND: SOURCE → COUCHBASE



OUTBOUND: COUCHBASE → BUSINESS HUB — CONSISTENT ACROSS EVERY SOURCE

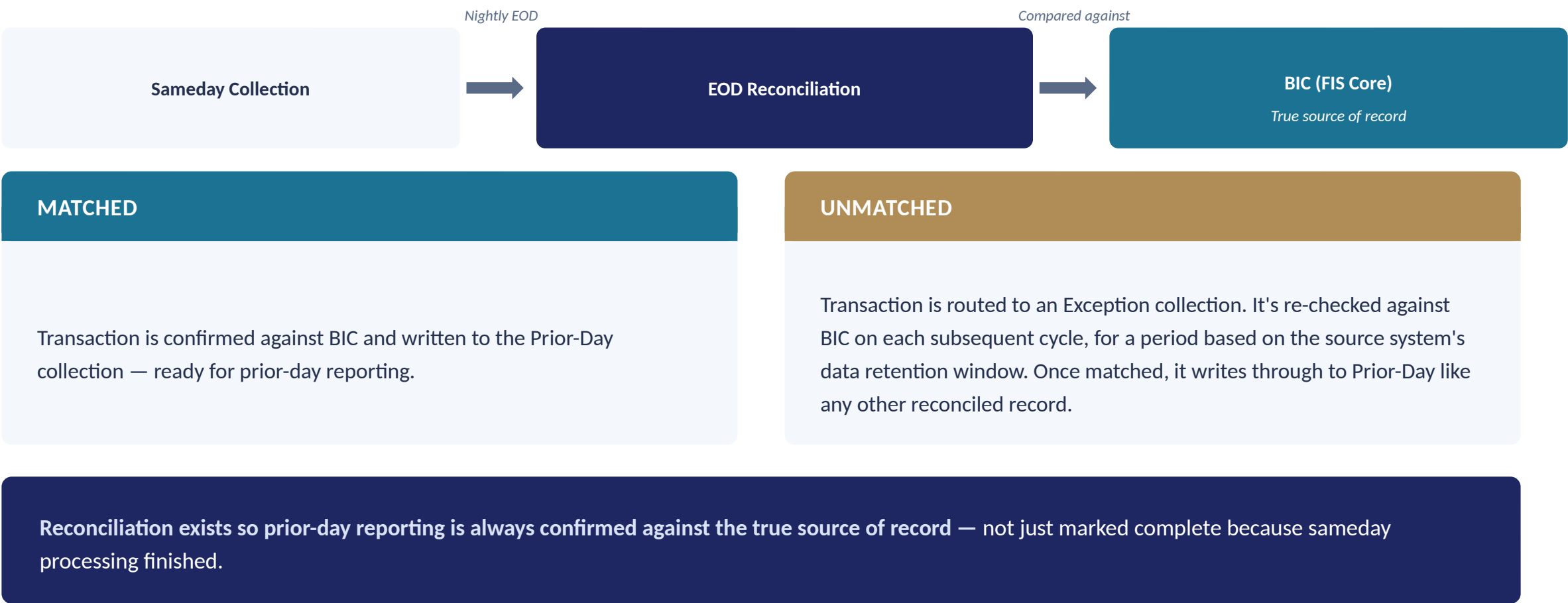


Applies to: IBS · RTP / FedNow · Wires · ACH · Lockbox

MuleSoft is the universal outbound gateway to Business Hub, and the direct inbound broker for Wires and ACH — the two sources that bypass Kafka entirely.

Reconciliation Process (Nightly)

Confirming sameday data against the true source of record before it becomes prior-day



Couchbase Platform: Cluster Overview

Cluster: PRDCCDMAZCEN_001 — Healthy — 13/27 nodes across 4 service groups

SERVICE GROUPS

Group	Nodes	CPU / RAM	Disk	IOPS	Disk Type
Data	5	16 vCPU / 64GB	1024GB	5000	P30
Index	3	16 vCPU / 64GB	256GB	1100	P15
Query	3	16 vCPU / 64GB	64GB	240	P6
Eventing	2	8 vCPU / 32GB	128GB	500	P10

BUCKET ARCHITECTURE

Bucket	Purpose	Scale
landing	Raw ingested data — batch EOD files, intraday TPS/MQ events	1.1M docs
process	Enriched & reconciled data — BAI detail/status/summary, prior-day & same-day	630K docs
semantic	Reporting-ready data — scheduled, standard, user-defined reports for Business Hub	—
history	7-day rolling reference copy of landing & process data, for operational lookback	13.9M docs
audit	Audit trail records	1.1M docs

The history bucket holds a 7-day reference window of landing/process data for operational lookback — it is not the 7-year archive. Long-term historical reporting is served from the Wintrust Data Warehouse via nightly EOD loads, not directly from Couchbase.

Landing & Process: Scopes & Collections

How raw data becomes reconciled, BAI-enriched data

LANDING

batch_eod scope

- ach_origination_indv_to_settlement
- ach_return
- tran_cms_detail / tran_cms_status / tran_cms_summary
- tran_deposit_detail / tran_deposit_status / tran_deposit_status_hist / tran_deposit_summary

intraday scope

- tps_events_deposits
- tps_file_ach
- tps_file_lockbox
- tps_mq_opf
- tps_mq_wires

PROCESS

batch_eod scope

- recon_matched
- tran_bai_detail_priorday
- tran_bai_status_priorday
- tran_bai_summary_priorday

intraday scope

- ach_intraday_match_queued
- tran_bai_detail_sameday
- tran_bai_status_sameday
- tran_bai_summary_sameday

Naming shows the transformation: **tran_deposit** (landing) becomes **tran_bai** (process) — enriched to the BAI code level — split into **priorday** and **sameday** collections matching the client's reporting requirement.

Semantic: Reporting Collections

What Business Hub actually queries

SCHEDULED REPORTS

1 collection

Pre-built recurring reports delivered on a fixed schedule.

STANDARD REPORTS

6 collections

Out-of-the-box reports covering common commercial banking needs.

USER-DEFINED REPORTS

3 collections

Custom reports configured to specific client or admin requirements.

The semantic bucket is what Business Hub actually queries — standard and custom reports built directly from the enriched process-layer data.

Business Hub Reports

What clients see, and when they see it

REPORT TYPES

Balance Position (BP)	User-Defined (UD)	Daily Statement (DS)
Client balance and position reporting	Custom, client-configured reports	Daily transaction statements

DATA TIMING — SHARED BY ALL THREE REPORT TYPES

Intraday / Sameday Served from the sameday collection as transactions arrive (e.g., balance_position_sameday).	End of Day / Prior Day Once EOD reconciliation completes, sameday data rolls into the prior-day collection (e.g., balance_position_priorday).
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SHARED USE-CASE FILTERS (BP, UD & DS)

Prior Month	Prior Day	Month-to-Date	Date Range
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SEARCH — A SEPARATE CAPABILITY

Clients search by amount, description, reference number, bank name, and more — against either sameday or prior-day data.

Whether it's a standard report or a free-form search, Couchbase serves every use case through prepared statements — one consistent retrieval pattern across Balance Position, User-Defined, Daily Statement, and Search.

Why Couchbase?

Not theoretical capability — the actual mechanics behind DataHub today

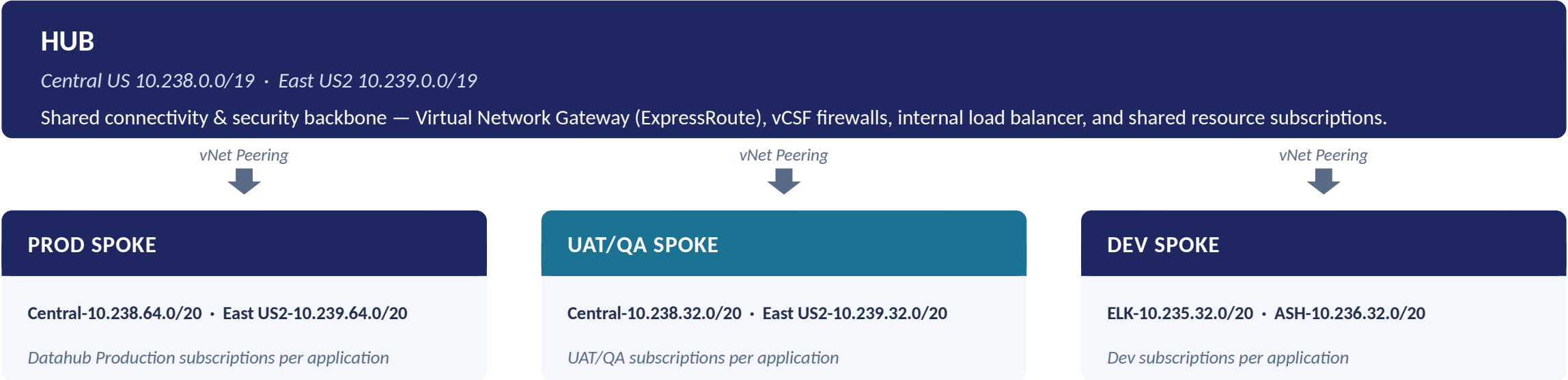
- Flexible Data Model: ingests IBS deposit JSON events, RTP/Wires XML messages, and ACH/Lockbox batch files into one document store — no rigid upfront schema across formats.
- Low-Latency Real-Time Access: powers Business Hub's prior-day/same-day balance visibility, sourced from Kafka-streamed IBS/RTP events filtered to commercial accounts.
- Unified Query + Search (N1QL): drives BAI-code level reporting from the process bucket (tran_bai_detail/status/summary, priorday/sameday) into the semantic bucket's standard, scheduled, and user-defined reports.
- Independently Scalable Services: Data / Index / Query / Eventing service groups scale separately as commercial transaction volume grows (13/27 nodes currently provisioned).
- Multi-Model, Single Platform: the landing → process → semantic lifecycle replaces what would otherwise be a separate staging DB, cache, and search engine.
- Built-in XDCR for Resilience: continuous one-way replication to East US2 gives Large Commercial/Corporate client data regional resilience.

Every one of these is already load-bearing in this deck — the actual mechanics of how DataHub ingests, enriches, and serves commercial banking data today.

Networking: Hub-and-Spoke Model

A shared backbone in the middle, isolated environments around it

On-Premises Data Centers (ELK-DC & ASH-DC) — connected via redundant 10G-LR ExpressRoute circuits



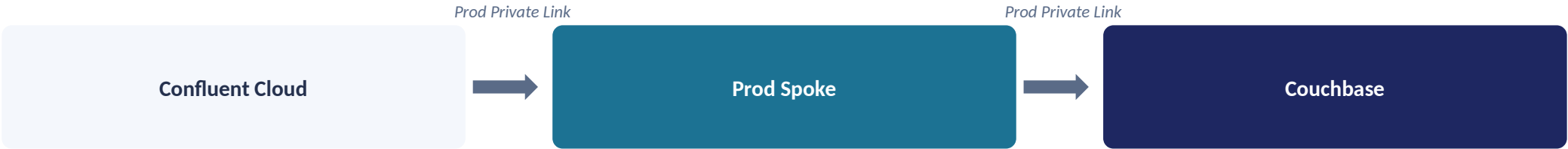
Spokes never connect directly to each other — only to the Hub.

The hub-spoke model centralizes connectivity and security once in the Hub, while keeping Prod, UAT/QA, and Dev fully isolated from each other — each spoke only talks to the Hub, never directly to another spoke.

How DataHub Communicates: A Real Example

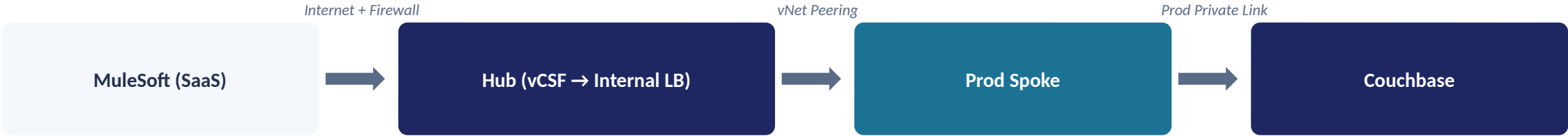
Private Link services connect directly to the spoke — only internet traffic routes through the Hub

PATH A: CONFLUENT → COUCHBASE (Private Link — direct to spoke)



Bypasses the Hub entirely — each environment (Prod / UAT-QA / Dev) has its own dedicated Private Link straight into that spoke.

PATH B: MULESOFT → COUCHBASE (Internet — via the Hub)



MuleSoft's SaaS traffic has no Private Link, so it enters over the internet through the Hub's vCSF firewall and Internal LB, then peers into the Prod spoke — a different path than Confluent's.

Private-Link services (Couchbase, Confluent, Informatica) connect straight into the relevant spoke. Only internet-sourced traffic like MuleSoft's is firewalled and routed through the Hub first. Either way, nothing reaches Couchbase directly from the public internet.

Private Link & Environment Addressing

Couchbase, Confluent, and Informatica each connect into the Hub via dedicated Private Link endpoints per environment (Prod / UAT-QA / Dev) — traffic never touches the public internet.

Environment	DataHub (ELK)	DataHub (ASH)	IDMC (ELK)	IDMC (ASH)
PROD	10.238.64.0/26	10.239.64.0/26	10.238.78.0/26	10.239.78.0/26
UAT/QA	10.238.32.0/26	10.239.32.0/26	10.238.46.0/26	10.239.46.0/26
DEV	10.238.48.0/24	10.239.48.0/24	10.235.32.0/27	10.236.32.0/27

PRIVATE LINK ENDPOINTS BY SOURCE SYSTEM

- Couchbase: one dedicated endpoint per environment (Prod / QA-UAT / Dev)
- Confluent: Prod, QA/UAT, and Dev Private Link
- Informatica (IDMC): Prod, QA/UAT, and Dev Private Link

HOW IPs MAP TO PRIVATE LINK

- Each subnet above hosts the Private Endpoint (NIC) for that service — Azure assigns it a private IP from that range.
- Private DNS resolves the service's hostname to that private IP, but only for traffic originating inside the VNet.
- ELK and ASH columns exist because Private Endpoints are regional — one per XDCR region.

This is the secure backbone underneath everything covered so far — every source-to-Couchbase and Couchbase-to-Business Hub path in this deck rides on this Private Link + ExpressRoute foundation.

Why This Architecture Matters

What breaks down without it

WITHOUT HUB-SPOKE

- Every spoke needs its own direct connection to on-prem and every shared service — connection sprawl instead of one shared backbone
- No natural isolation — a Dev misconfiguration could expose a path toward Prod
- Security policy duplicated per environment instead of centralized once in the Hub

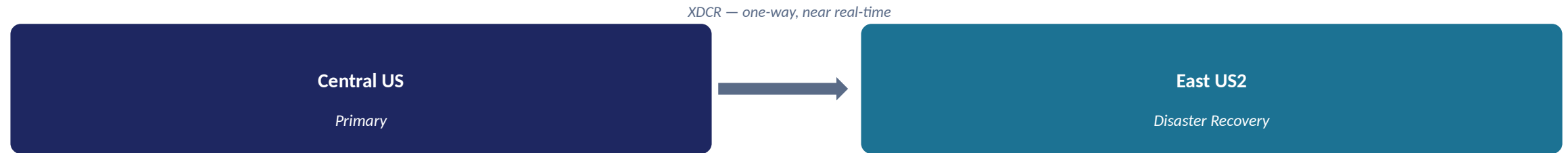
WITHOUT PRIVATE LINK

- Confluent / Couchbase / Informatica traffic would traverse the public internet to reach the Hub
- Commercial banking data exposed to internet attack surface
- Harder to meet compliance expectations for private, non-internet-routable financial data paths

Hub-and-spoke plus Private Link together mean commercial banking data never touches the public internet, and every environment is isolated by design — not by convention.

Disaster Recovery: Regional Failover

Primary in Central US, continuously mirrored to East US2



HOW FAILOVER WORKS

- Data copies over to East US2 in near real-time, using XDCR. It's one-way and always running — no manual step needed.
- If Central US goes down, we don't flip over automatically. We follow a runbook and switch over on purpose, in a controlled way.
- Full RTO/RPO numbers and failover steps live in a separate runbook ([link goes here](#)).

Because East US2 already has a live copy of the data, switching over means promoting it — not rebuilding it. That's what keeps downtime low.

Future Use Case: Encompass Webhooks

PROPOSED — DESIGN NOT FINAL YET

Why Couchbase fits — not a finalized design

Quick flow: Encompass sends a webhook → MuleSoft/Kafka → lands in a new Couchbase bucket.

WHY COUCHBASE FITS

- A webhook only tells us what changed — like a loan status or an address — not the whole loan.
- Couchbase can update just that one field directly instead of reading and rewriting the whole document. This is called a subdocument mutation.
- That matters here because these events come in fast, and this saves us from doing more work than we need to.

Still just an idea we're testing, not a final design — but subdoc mutations are why Couchbase looks like a strong fit here.

Future Use Case: Mittera

PROPOSED — DESIGN NOT FINAL YET

Why Couchbase fits — not a finalized design

Quick flow: the data Mittera needs already sits in Couchbase — nothing new to build to get it there.

WHY COUCHBASE FITS

- We already keep this data merged in Couchbase.
- All Mittera needs is one pull a day, for the prior day's data.
- We already have solid indexes built for that kind of pull, so it stays fast.

Not a new pipeline. Just a new place reading data we already have.

Future Use Case: IDMC

PROPOSED — DESIGN NOT FINAL YET

Why Couchbase fits — not a finalized design

Quick flow: IDMC events come into Couchbase and get processed as MDM data.

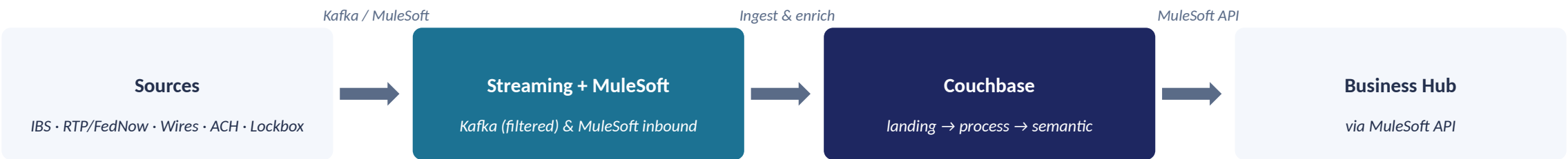
WHY COUCHBASE FITS

- The private network connection for IDMC is already set up — nothing new to build there either.
- Couchbase doesn't need a fixed schema, so it can handle different MDM record shapes without a problem.

Same connection, same flexibility we're already using elsewhere in this deck — just a new use for it.

End-to-End Flow Recap

The full picture — from source to Business Hub, and who owns what



All data in motion travels through the Hub-and-Spoke network via Private Link — see Networking.

TEAM OWNERSHIP

BUSINESS HUB

Built & operated by FIS — the client-facing portal for balances, transactions, reports, and search.

DATAHUB

Built & operated by Wintrust Enterprise DM — acquires, streams, integrates, and persists data in Couchbase for Business Hub to consume.

From five source systems to one client-facing portal: filtered, enriched, and reconciled through Couchbase, delivered through MuleSoft, and secured end-to-end by the Hub-and-Spoke network — this is DataHub.